



Part I

Lab Assignment-5 (Introduction to Object-Oriented Programming)

Department: CSE
(22CPP209)

Subject: OOAD Lab

Objective: Understanding *const* Objects and Member functions and Basic implementation of *Smart pointers*.

Instruction: Modify the previous code to explore the calling mechanism of *const* objects and member functions. The implementation should use the smart pointer to manage the dynamic memory/object allocation.

Question 1. Create a class *Student* with data members for student name, roll number, and marks. Implement member functions to input and display student information. Demonstrate the concepts of shallow and deep copy by implementing a copy constructor and an assignment operator.

Question 2: Design a C++ program to simulate a bank account management system using object-oriented programming principles. Implement a **BankAccount** class with constructors, including a copy constructor, and a copy assignment operator. The class contains attributes like account number, account holder, and balance, and include member functions for depositing, withdrawing, and displaying account details. Demonstrate different scenarios by creating, copying, and assigning objects of these classes, and performing various operations on them.